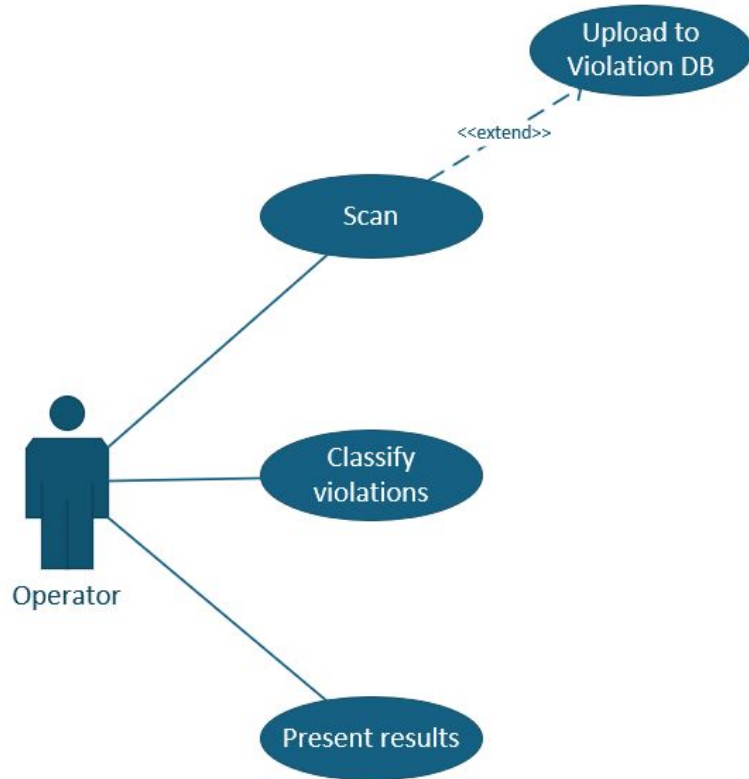


Static Code Analysis Framework

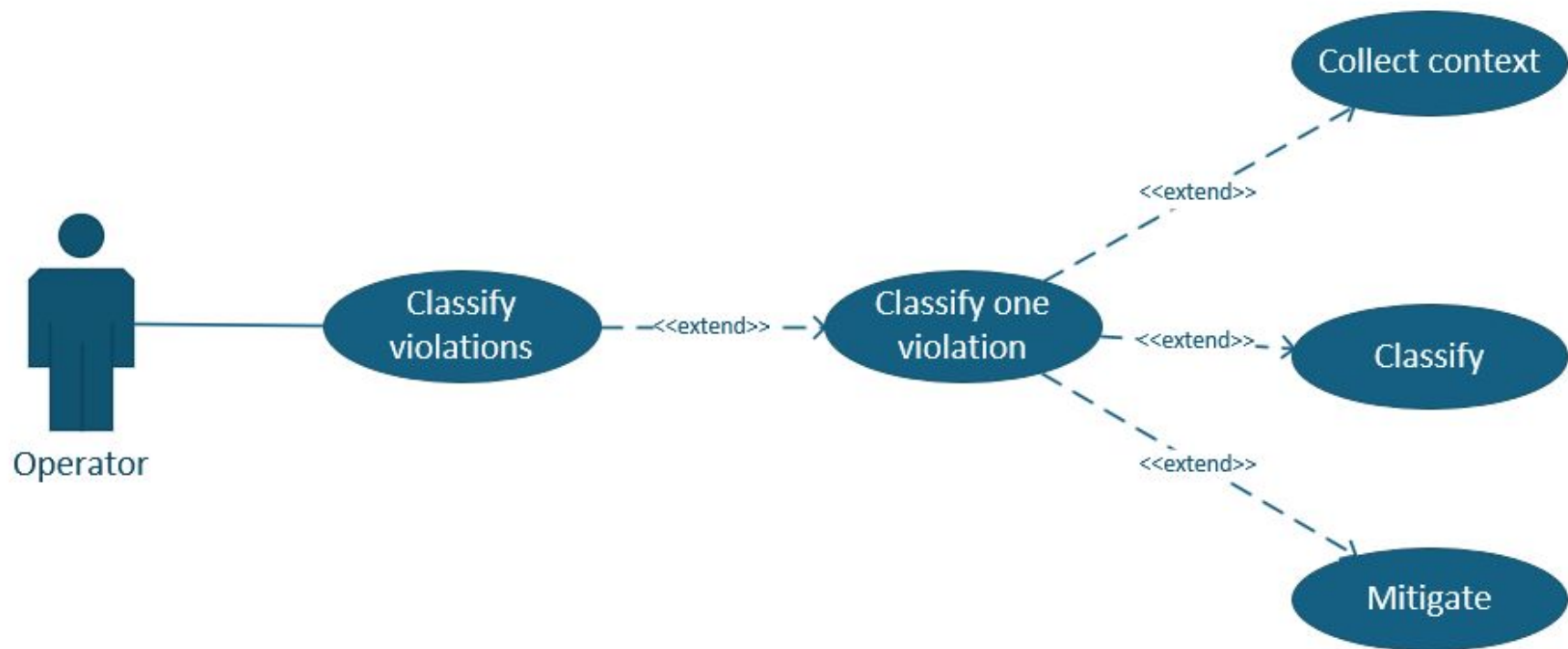
Architectural design

ELISA Tools WG

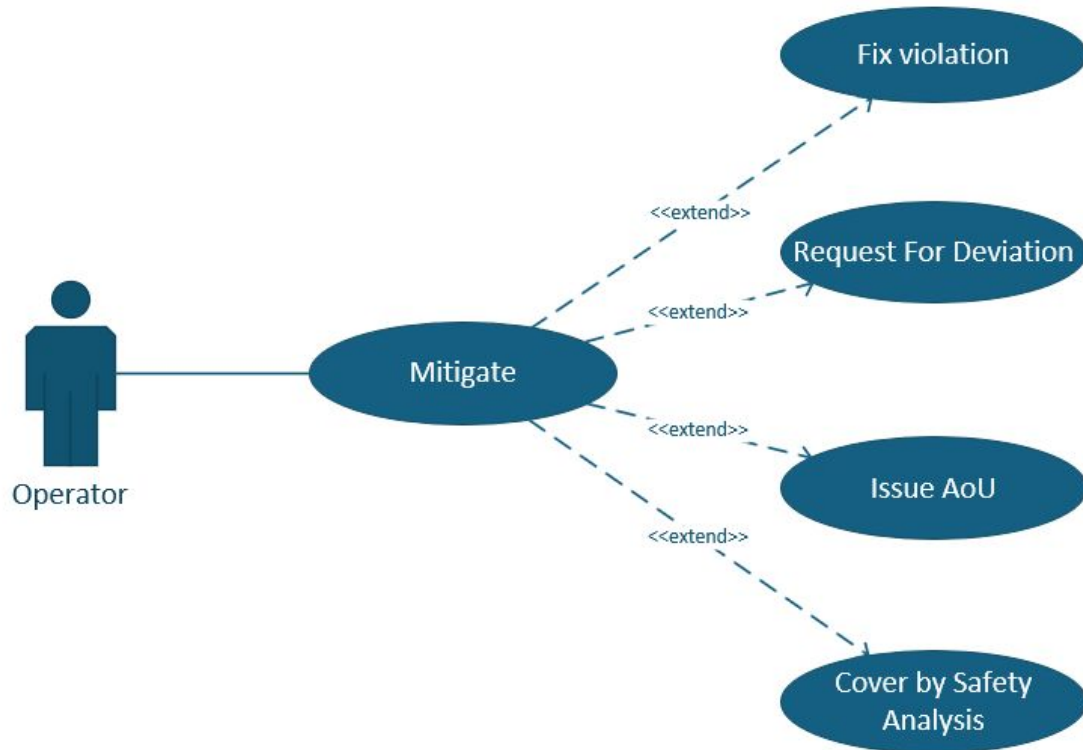
Use cases - high level



Use case - classification



Use case - mitigation

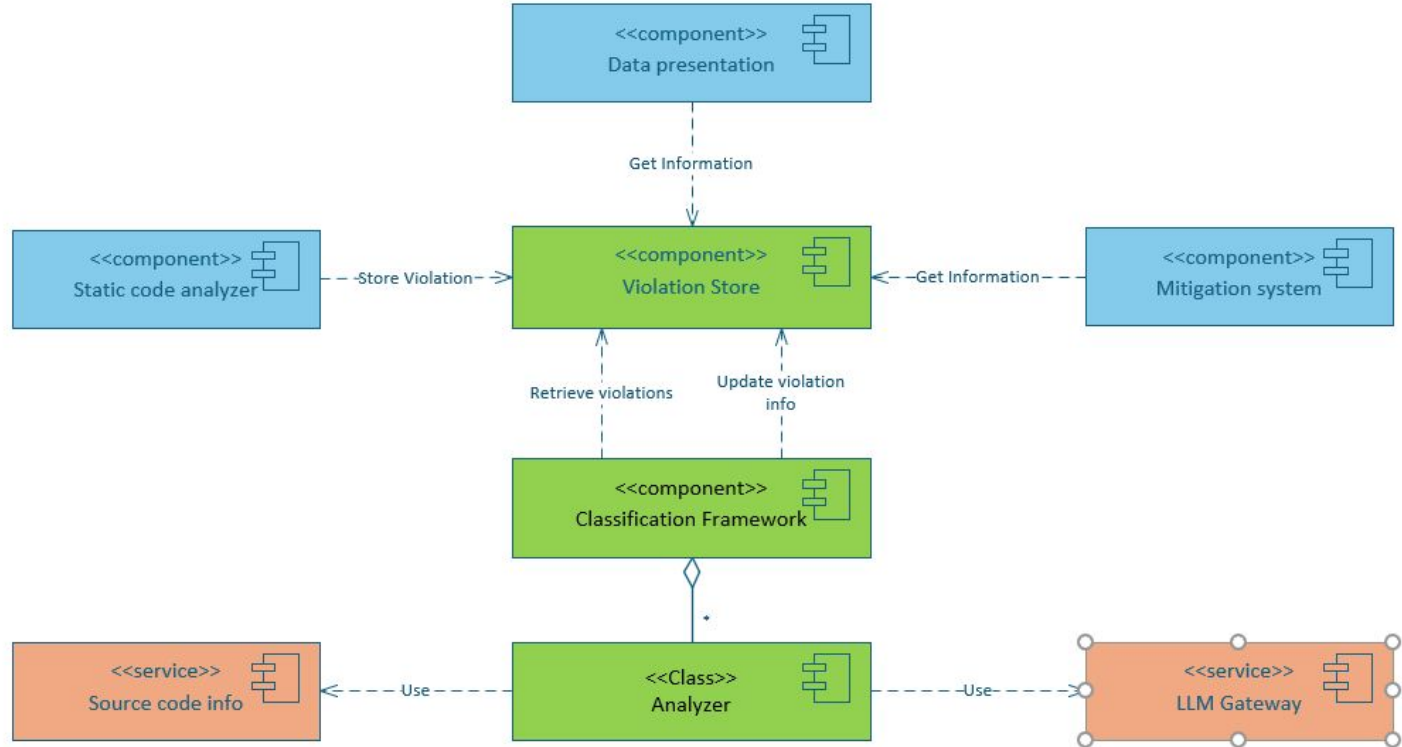


High level architecture and integration points

Green components are the core Static Code Analysis System

Blue components are third party components that integrate with the system

Orange components are services that are used by the system



Violation store

- Implemented as a MongoDB
- Normalizes violations to the following format
 - Values hidden to avoid legal issues
- Classification uses the violations in the ViolationDB

```
_id: ObjectId('6a99e912bac3960f1fced5f9')
file_path : 
line_number : 15
char_index : 1
rule_family : 
rule_type : "Rule"
rule_number : 
rule_category : 
▼ text : Array (2)
  0: "/workspace
  1: " 1.
```

Service: Source code info

- Any package that implements the source-info API
 - Preferred: KSNNav
 - Optional:
 - In-house tool
 - Cscope
- Sample API functions:
 - `get_function_body_by_name`
 - `get_macro_by_name`
 - `get_type_definition_by_name`
 - ...

Service: LLM gateway

- A service that provides access to an LLM
- Example:
 - build.nvidia.com
 - <http://tokenfactory.nebius.com>